City Information Facts

city_info(City, Country, Area, Budget, Climate, Food)

This predicate is used to define facts about different cities. Each fact contains information such as the city name (`City`), country (`Country`), area (`Area`), budget level (`Budget`), climate (`Climate`), and food preferences (`Food`).

Recommendation Rule

recommend_package(City, Country, Area, Budget, Climate, Food) :-

city_info(City, Country, Area, Budget, Climate, Food).

Explanation:

This rule serves as the recommendation mechanism. It states that a travel package is recommended if there exists a city information fact matching the specified criteria (country, area, budget, climate, and food preferences).

User Interaction

interact:-

writeln("Welcome to the Travel Package Recommendation System!"),

check_input("country", Country),

check_input("area", Area),

check_input("budget", Budget),

check_input("climate", Climate),

check_input("food", Food),

recommend_package(City, Country, Area, Budget, Climate, Food),

write("Recommended Package: "), write(City), nl.

Explanation:

The `interact` predicate initiates the interaction with the user. It prompts the user to input preferences for country, area, budget, climate, and food using the `check_input` predicate. After obtaining user preferences, it calls the `recommend_package` rule to recommend a travel package and displays the result.

Input Verification

check_input(Preference, Value) :-

writeln("Enter your preferred " -Preference),

read(Value),

```
(verify_input(Preference, Value) ->
    true
;   writeln("Invalid input. Please try again."), check_input(Preference, NewValue)
).
```

Explanation:

The `check_input` predicate is responsible for taking user input, verifying its validity using the `verify_input` predicate, and prompting the user to try again if the input is invalid.

Input Verification

```
verify_input(Preference, Value) :-
city_info(_, Country, Area, Budget, Climate, Food),
((Preference = "country", Value = Country);
(Preference = "area", Value = Area);
(Preference = "budget", Value = Budget);
(Preference = "climate", Value = Climate);
(Preference = "food", Value = Food)).
```

Explanation:

The `verify_input` predicate checks whether the user's input (`Value`) matches any of the information available in the city facts for the specified preference (`Preference`). It ensures that the input is valid based on the predefined city information.